Math 54 — Week 3

Linear independence; linear transformations; the matrix of a linear transformation (Lay 1.7–1.9)

§1 Facts used from Weeks 1–2

(W1) A consistent system has exactly one solution if it has no free variables, infinitely many otherwise. Free variables correspond to non-pivot columns.

(W2) $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$, where $\mathbf{a}_j$ are the columns of A. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is a linear combination of the columns (Lay 1.4, Thm. 3).

(W3) The columns of an $m \times n$ matrix A span $\mathbb{R}^m$ exactly when A has a pivot in every row (Lay 1.4, Thm. 4).

(W4) $A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ and $A(c\mathbf{u}) = c(A\mathbf{u})$ (Lay 1.4, Thm. 5).

(W5) $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution exactly when it has a free variable.

(W6) Each pivot lies in a different row and a different column, so an $m \times n$ matrix has at most m and at most n pivots. For a square matrix, a pivot in every column is the same as a pivot in every row.

§2 This week

§2.1 Linear independence (Lay 1.7)

The question. In Week 2 the vectors $\mathbf{a}_1 = (1,-2,3)$, $\mathbf{a}_2 = (0,1,2)$, $\mathbf{a}_3 = (2,-5,4)$ satisfied $\mathbf{a}_3 = 2\mathbf{a}_1 - \mathbf{a}_2$, so $\mathbf{a}_3$ added nothing to the span of $\mathbf{a}_1, \mathbf{a}_2$. Is one of $(1,2,3)$, $(4,5,6)$, $(2,1,0)$ a combination of the others, and how do we detect such redundancy in general?

Definition. A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation $x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$ has only the trivial solution. It is **linearly dependent** if there are weights $c_1, \ldots, c_p$, *not all zero*, with $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$; such an equation is a **linear dependence relation**.

Columns of a matrix. By (W2), the equation $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ is $A\mathbf{x} = \mathbf{0}$. So:

1. the columns of A are linearly independent exactly when $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
2. by (W5), that happens exactly when $A\mathbf{x} = \mathbf{0}$ has no free variable;
3. that is, exactly when A has **a pivot in every column**.

Worked example. Let $\mathbf{v}_1 = (1,2,3)$, $\mathbf{v}_2 = (4,5,6)$, $\mathbf{v}_3 = (2,1,0)$. Row reduce the matrix with these columns (the zero right-hand side stays zero):

$$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 6 & 0 \end{bmatrix} \xrightarrow[R_3-3R_1]{R_2-2R_1} \begin{bmatrix} 1 & 4 & 2 \\ 0 & -3 & -3 \\ 0 & -6 & -6 \end{bmatrix} \xrightarrow{R_3-2R_2} \begin{bmatrix} 1 & 4 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_1-4R_2]{-\frac{1}{3}R_2} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

1. Column 3 has no pivot, so x_3 is free and the set is dependent.

2. The rows say $x_1 = 2x_3$ and $x_2 = -x_3$. Taking $x_3 = 1$ gives $x_1 = 2$, $x_2 = -1$, so $2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$.
3. Check: $(2,4,6) - (4,5,6) + (2,1,0) = (0,0,0)$.

Small sets.

- **One vector**: $\{\mathbf{v}\}$ is independent exactly when $\mathbf{v} \neq \mathbf{0}$. Reason: if $\mathbf{v} \neq \mathbf{0}$, some entry $v_i \neq 0$, and $x\mathbf{v} = \mathbf{0}$ forces $xv_i = 0$, so $x = 0$; if $\mathbf{v} = \mathbf{0}$, then $1 \cdot \mathbf{v} = \mathbf{0}$ is a dependence relation.
- **Two vectors**: $\{\mathbf{v}_1, \mathbf{v}_2\}$ is dependent exactly when one is a multiple of the other.
  1. **Assume one is a multiple of the other**, say $\mathbf{v}_1 = c\mathbf{v}_2$. Then $1 \cdot \mathbf{v}_1 - c\mathbf{v}_2 = \mathbf{0}$ is a dependence relation (weight $1 \neq 0$). The case $\mathbf{v}_2 = c\mathbf{v}_1$ is the same with the roles exchanged: $-c\mathbf{v}_1 + 1 \cdot \mathbf{v}_2 = \mathbf{0}$.
  2. **Assume the set is dependent**: $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{0}$ with c_1, c_2 not both 0. If $c_1 \neq 0$, then $\mathbf{v}_1 = -\frac{c_2}{c_1}\mathbf{v}_2$. If $c_1 = 0$, then $c_2 \neq 0$ and $c_2\mathbf{v}_2 = \mathbf{0}$, so $\mathbf{v}_2 = \mathbf{0} = 0 \cdot \mathbf{v}_1$. Either way one vector is a multiple of the other.

  Geometrically, two vectors are dependent exactly when they lie on one line through $\mathbf{0}$ (left panel of the figure).

Theorem (Characterization of linearly dependent sets (Lay 1.7, Thm. 7))
A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one vector in S is a linear combination of the others. In fact, if S is dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ with $j > 1$ is a linear combination of the *preceding* vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

The idea

In a dependence relation some vector has a nonzero weight, and an equation can be solved for any variable whose coefficient is nonzero. To get the stronger statement, solve for the *last* vector with a nonzero weight: every other vector in the relation comes before it.

Proof. 1. **Assume some $\mathbf{v}_j$ is a combination of the others**, $\mathbf{v}_j = \sum_{i \neq j} c_i\mathbf{v}_i$. Then $\sum_{i \neq j} c_i\mathbf{v}_i - \mathbf{v}_j = \mathbf{0}$ is a relation in which $\mathbf{v}_j$ has weight $-1 \neq 0$. So S is dependent.

2. **Assume S is dependent**: $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ with some $c_j \neq 0$. Solving for $\mathbf{v}_j$ gives

$$\mathbf{v}_j = -\frac{1}{c_j}\sum_{i \neq j} c_i\mathbf{v}_i,$$

a combination of the others. (If c_j is the only nonzero weight, this says $\mathbf{v}_j = \mathbf{0}$, the combination with all weights 0.)

3. **Assume moreover $\mathbf{v}_1 \neq \mathbf{0}$.** Choose j to be the *largest* index with $c_j \neq 0$, so the relation is $c_1\mathbf{v}_1 + \cdots + c_j\mathbf{v}_j = \mathbf{0}$. If $j = 1$ it says $c_1\mathbf{v}_1 = \mathbf{0}$ with $c_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$, which is false. So $j > 1$, and

$$\mathbf{v}_j = -\frac{1}{c_j}\big(c_1\mathbf{v}_1 + \cdots + c_{j-1}\mathbf{v}_{j-1}\big).$$

□

Theorem (Too many vectors are dependent (Lay 1.7, Thm. 8))
If $p > n$, any set of p vectors in $\mathbb{R}^n$ is linearly dependent.

Proof. 1. The matrix $A = [\,\mathbf{v}_1 \cdots \mathbf{v}_p\,]$ is $n \times p$, so it has at most n pivots (W6).

2. It has $p > n$ columns, so some column has no pivot: $A\mathbf{x} = \mathbf{0}$ has a free variable, hence a nontrivial solution (W5).

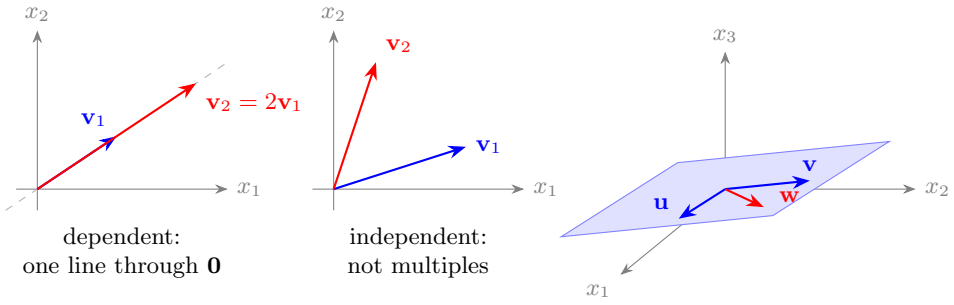
□

Theorem (A set containing **0** is dependent (Lay 1.7, Thm. 9))
If a set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ contains the zero vector, it is linearly dependent.

Proof. Renumber so that $\mathbf{v}_1 = \mathbf{0}$. Then $1 \cdot \mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_p = \mathbf{0}$ is a dependence relation, because the weight 1 is not zero. □

Adding and removing vectors.

- A set that contains a dependent set is dependent: use the same relation, with weight 0 on the added vectors. So every subset of an independent set is independent: a dependent subset would make the whole set dependent.
- **Assume $\mathbf{u}, \mathbf{v}$ are independent.** Then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is dependent exactly when $\mathbf{w}$ is in $\mathrm{Span}\{\mathbf{u}, \mathbf{v}\}$. If $\mathbf{w}$ is in the span, the set is dependent by Theorem 7. Conversely, take a relation $c_1\mathbf{u} + c_2\mathbf{v} + c_3\mathbf{w} = \mathbf{0}$, not all weights 0. If $c_3 = 0$, then $c_1\mathbf{u} + c_2\mathbf{v} = \mathbf{0}$ with c_1, c_2 not both 0, contradicting the independence of $\mathbf{u}, \mathbf{v}$. So $c_3 \neq 0$, and solving for $\mathbf{w}$ gives
$$\mathbf{w} = -\frac{c_1}{c_3}\mathbf{u} - \frac{c_2}{c_3}\mathbf{v}.$$
- **Geometry in $\mathbb{R}^3$.** If $\mathbf{u}, \mathbf{v}$ are not multiples of each other, they are independent and span a plane through $\mathbf{0}$; by the previous item, $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is dependent exactly when $\mathbf{w}$ lies in that plane (right panel).



Left and middle: two vectors are dependent exactly when they lie on one line through **0**. Right: three vectors, the first two independent, are dependent exactly when the third lies in the plane spanned by the first two.

The question answered. The worked example gives $2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, so $(2, 1, 0) = (4, 5, 6) - 2(1, 2, 3)$ is a combination of the others. In general, put the vectors into the columns of a matrix and row reduce: the set is independent exactly when every column has a pivot, and a free variable gives a dependence relation.

Shortcuts

- **Independence by inspection**: a set with more vectors than entries (Thm. 8), or containing **0** (Thm. 9), is dependent; for two vectors, check whether one is a multiple of the other. Instance: $(1, 0)$, $(0, 1)$, $(2, 3)$ in $\mathbb{R}^2$ are dependent (three vectors, two entries). Otherwise row reduce $[\,\mathbf{v}_1 \cdots \mathbf{v}_p\,]$ and look for a pivot in every column.
- **Finding a dependence relation**: solve $A\mathbf{x} = \mathbf{0}$, set one free variable to 1 and the others to 0. Reason: the resulting solution has a nonzero entry, so its entries are weights of a dependence relation. Instance: $x_3 = 1$ in the worked example.
- **A column relation is a solution of $A\mathbf{x} = \mathbf{0}$**: if $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$, then $\mathbf{x} = (c_1, \ldots, c_n)$ solves $A\mathbf{x} = \mathbf{0}$. Reason: (W2). Instance: $\mathbf{a}_1 + 2\mathbf{a}_2 = \mathbf{a}_3$ means $\mathbf{a}_1 + 2\mathbf{a}_2 - \mathbf{a}_3 = \mathbf{0}$, so $\mathbf{x} = (1, 2, -1)$.

Traps

- Three vectors in $\mathbb{R}^3$, no two of which are multiples, can still be dependent: the vectors of the worked example lie in one plane.

- In a dependent set, not every vector need be a combination of the others: in $\{(1,0),(2,0),(0,1)\}$, $(0,1)$ is not.
- One vector that is not a combination of the others does not make a set independent. In $\{(1,0),(2,0),(0,1),(0,0)\}$, the vector $(0,1)$ is not a combination of the other three, but the set contains $\mathbf{0}$.
- Theorem 8 says nothing when $p \le n$: fewer vectors than entries can still be dependent, as $(1,2,0)$ and $(2,4,0)$ in $\mathbb{R}^3$ show.
- Independence concerns $A\mathbf{x} = \mathbf{0}$, not whether $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$. The columns $(1,0,0)$ and $(0,1,0)$ are independent, yet $A\mathbf{x} = (0,0,1)$ has no solution.

§2.2 Linear transformations (Lay 1.8)

The question. Instead of solving $A\mathbf{x} = \mathbf{b}$ for one $\mathbf{b}$, think of $\mathbf{x} \mapsto A\mathbf{x}$ as a function that moves every vector of $\mathbb{R}^n$. What does the matrix below do to the unit square with corners $(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$, and which properties do all such functions share?

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

Definition. A **transformation** (or function) $T\colon \mathbb{R}^n \to \mathbb{R}^m$ assigns to each $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$, the **image** of $\mathbf{x}$. $\mathbb{R}^n$ is the **domain** and $\mathbb{R}^m$ the **codomain**; the set of all images $T(\mathbf{x})$ is the **range**. A **matrix transformation** is $T(\mathbf{x}) = A\mathbf{x}$ for an $m \times n$ matrix A.

- The range of $\mathbf{x} \mapsto A\mathbf{x}$ is the span of the columns of A, because by (W2) the vectors $A\mathbf{x}$ are exactly the linear combinations of the columns.
- The **standard vectors** of $\mathbb{R}^n$ are the columns $\mathbf{e}_1, \dots, \mathbf{e}_n$ of I_n: $\mathbf{e}_j$ has entry 1 in position j and 0 elsewhere. Every $\mathbf{x}$ in $\mathbb{R}^n$ is $x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, because entry i of the right side is x_i.
- For $T(\mathbf{x}) = A\mathbf{x}$, "is $\mathbf{b}$ in the range of T?" is an *existence* question (is $A\mathbf{x} = \mathbf{b}$ consistent?), and "is there only one $\mathbf{x}$ with $T(\mathbf{x}) = \mathbf{b}$?" is a *uniqueness* question.

Worked example: range and uniqueness. Let $T(\mathbf{x}) = A\mathbf{x}$ with A below, and take $\mathbf{b} = (4,1,7)$ and $\mathbf{c} = (1,1,0)$. Apply $R_3 - 2R_1$, then $R_3 + R_2$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 3 \end{bmatrix}, \qquad \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 2 & 3 & 7 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right], \qquad \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 3 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{array}\right].$$

1. $\mathbf{b}$ is in the range: $x_2 = 1$, $x_1 = 4 - 2x_2 = 2$, so $T(2,1) = \mathbf{b}$. There is no free variable, so $(2,1)$ is the only such $\mathbf{x}$.
2. $\mathbf{c}$ is not in the range: the last row is $0 = -1$.

Definition. T is **linear** if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for all $\mathbf{u}, \mathbf{v}$ in the domain and all scalars c.

Every matrix transformation is linear, by (W4).

Theorem (Properties of linear transformations (Lay 1.8))
If T is linear, then $T(\mathbf{0}) = \mathbf{0}$ and $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ and scalars c, d. Conversely, a transformation with $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}, c, d$ is linear.

Proof. 1. **Assume T is linear.** $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}$. Also $T(c\mathbf{u} + d\mathbf{v}) = T(c\mathbf{u}) + T(d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$, by the two rules in turn.

2. **Assume $T(c\mathbf{u}+d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ for all choices.** With $c = d = 1$ this is $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$. With $d = 0$ and $\mathbf{v} = \mathbf{0}$ it is $T(c\mathbf{u}) = cT(\mathbf{u}) + 0\,T(\mathbf{0}) = cT(\mathbf{u})$.

□

Consequences.

- **Superposition.** Repeating the two rules gives, for any vectors and weights,

$$T(c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p) = c_1T(\mathbf{v}_1) + \cdots + c_pT(\mathbf{v}_p).$$

- **Dependent sets go to dependent sets.** If $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ with weights not all 0, then by superposition $c_1T(\mathbf{v}_1) + \cdots + c_pT(\mathbf{v}_p) = T(\mathbf{0}) = \mathbf{0}$ with the same weights.
- **Lines go to lines or points.** By superposition, $T(\mathbf{p} + t\mathbf{v}) = T(\mathbf{p}) + tT(\mathbf{v})$. As t varies, this is the line through $T(\mathbf{p})$ parallel to $T(\mathbf{v})$ if $T(\mathbf{v}) \neq \mathbf{0}$, and the single point $T(\mathbf{p})$ if $T(\mathbf{v}) = \mathbf{0}$. In the same way the segment $\mathbf{p} + t(\mathbf{q} - \mathbf{p})$, $0 \le t \le 1$, goes to the segment from $T(\mathbf{p})$ to $T(\mathbf{q})$ (or to one point).

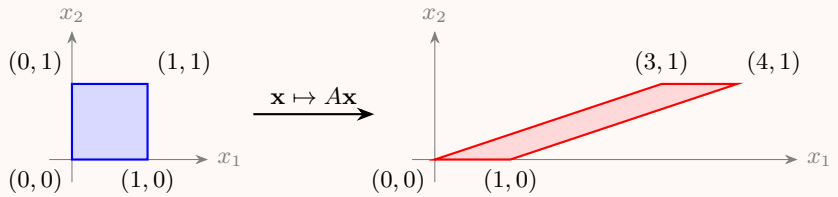
Examples on $\mathbb{R}^2$.

$\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$ shear

$\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$ projection onto the x_1-axis

$\begin{bmatrix}r & 0\\ 0 & r\end{bmatrix}$ contraction $(0 \le r \le 1)$, dilation $(r > 1)$

The shear sends (x_1, x_2) to $(x_1 + 3x_2,\ x_2)$: each point moves horizontally by 3 times its height. The projection sends (x_1, x_2) to $(x_1, 0)$. The matrix rI sends $\mathbf{x}$ to $r\mathbf{x}$.



The shear of the question. The corners go to $A(0,0) = (0,0)$, $A(1,0) = (1,0)$, $A(1,1) = (4,1)$, $A(0,1) = (3,1)$, and each edge goes to the segment between the images of its endpoints.

The question answered. The square goes to the parallelogram with corners $(0,0)$, $(1,0)$, $(4,1)$, $(3,1)$: the bottom edge stays fixed and the top edge slides 3 to the right. Every linear transformation, and in particular every matrix transformation, sends $\mathbf{0}$ to $\mathbf{0}$, sends combinations to the same combinations of the images, and sends lines and segments to lines and segments (or single points).

Shortcuts

- **Is T linear?** First check $T(\mathbf{0}) = \mathbf{0}$; if it fails, T is not linear (properties of linear transformations, Lay 1.8). Then test the two rules, or write T as $A\mathbf{x}$. Instance: $T(\mathbf{x}) = \mathbf{x} + (1,0)$ has $T(\mathbf{0}) = (1,0)$, so it is not linear.
- **Images of combinations**: given $T(\mathbf{u})$ and $T(\mathbf{v})$, $T(c\mathbf{u}+d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$, with no matrix needed. Reason: properties of linear transformations (Lay 1.8). Instance: $T(\mathbf{u}) = (1,2)$ and $T(\mathbf{v}) = (0,1)$ give $T(3\mathbf{u} - \mathbf{v}) = (3,6) - (0,1) = (3,5)$.
- **Image of a polygon**: map the corners and join them in the same order. Reason: segments go to segments. Instance: the figure.
- **Range of $\mathbf{x} \mapsto A\mathbf{x}$** is the span of the columns of A. Reason: (W2). Instance: the projection onto the x_1-axis has columns $(1,0)$ and $(0,0)$, so its range is the x_1-axis.

Traps

- $T(\mathbf{0}) = \mathbf{0}$ does not make T linear: $T(x_1, x_2) = (x_1^2, x_2)$ has $T(\mathbf{0}) = \mathbf{0}$, but $T(2\mathbf{e}_1) = (4, 0)$ while $2T(\mathbf{e}_1) = (2, 0)$.
- The range is a subset of the codomain and can be smaller: the projection onto the x_1-axis has codomain $\mathbb{R}^2$ and range a line.
- A line need not go to a line: the projection onto the x_1-axis sends the vertical line $x_1 = 1$ to the single point $(1, 0)$.

§2.3 The matrix of a linear transformation (Lay 1.9)

The question. Is every linear transformation $\mathbb{R}^n \to \mathbb{R}^m$ a matrix transformation? If so, how do we find the matrix from a description such as "rotate by 30°", and how do we tell from it whether every $\mathbf{b}$ is an image, and of only one $\mathbf{x}$?

Theorem (The standard matrix (Lay 1.9, Thm. 10))

Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be linear. Then there is exactly one matrix A with $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$, namely

$$A = [\,T(\mathbf{e}_1)\ \ T(\mathbf{e}_2)\ \ \cdots\ \ T(\mathbf{e}_n)\,].$$

A is the **standard matrix** of T.

The idea

Every vector is a combination of $\mathbf{e}_1, \ldots, \mathbf{e}_n$, and a linear map sends a combination to the same combination of images. So the n images $T(\mathbf{e}_j)$ determine T completely, and a matrix with those columns does the same job.

Proof. 1. Write $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. By superposition and then the definition of $A\mathbf{x}$,

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = A\mathbf{x}.$$

2. Uniqueness: if also $T(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x}$, then column j of B is $B\mathbf{e}_j = T(\mathbf{e}_j)$, which is column j of A.

□

Worked example: the matrix from a formula. For $T(x_1, x_2) = (x_1 - 2x_2,\ 3x_2,\ 4x_1 + x_2)$, the images of the standard vectors are $T(\mathbf{e}_1) = (1, 0, 4)$ and $T(\mathbf{e}_2) = (-2, 3, 1)$, so

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 4 & 1 \end{bmatrix}.$$

Row i of A lists the coefficients of x_1, x_2 in entry i of $T(\mathbf{x})$, as the row-vector rule requires. (This formula shows T is linear, since $T(\mathbf{x}) = A\mathbf{x}$.)

Rotation. Let T rotate $\mathbb{R}^2$ counterclockwise about $\mathbf{0}$ by the angle φ. Rotation is linear:

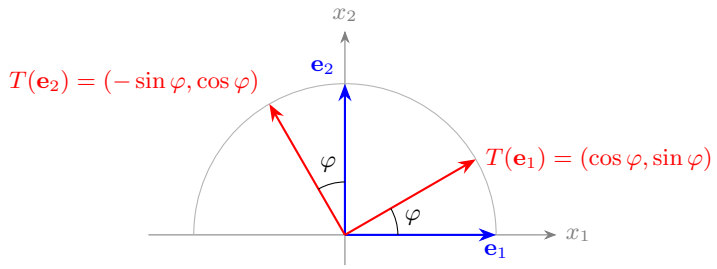
1. **Multiples.** T carries the line through $\mathbf{0}$ and $\mathbf{u}$ to the line through $\mathbf{0}$ and $T(\mathbf{u})$, keeps every distance from $\mathbf{0}$, and sends the half of the line on the side of $\mathbf{u}$ to the half on the side of $T(\mathbf{u})$ (and the opposite half to the opposite half). The point $c\mathbf{u}$ is $|c|$ times as far from $\mathbf{0}$ as $\mathbf{u}$, on the side of $\mathbf{u}$ if $c > 0$ and on the opposite side if $c < 0$; so its image is $|c|$ times as far from $\mathbf{0}$ as $T(\mathbf{u})$, on the matching side, which is $cT(\mathbf{u})$.
2. **Sums, u and v not on one line through 0.** T carries the parallelogram with sides $\mathbf{u}, \mathbf{v}$, whose fourth vertex is $\mathbf{u}+\mathbf{v}$, to the parallelogram with sides $T(\mathbf{u}), T(\mathbf{v})$, whose fourth vertex is $T(\mathbf{u})+T(\mathbf{v})$. So $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u})+T(\mathbf{v})$.

3. **Sums, u and v on one line through 0.** If $\mathbf{u} = \mathbf{0}$, both sides equal $T(\mathbf{v})$, since $T(\mathbf{0}) = T(0\mathbf{v}) = 0\,T(\mathbf{v}) = \mathbf{0}$ by step 1. Otherwise $\mathbf{v} = c\mathbf{u}$, and by step 1,

$$T(\mathbf{u}+\mathbf{v}) = T((1+c)\mathbf{u}) = (1+c)T(\mathbf{u}) = T(\mathbf{u}) + T(c\mathbf{u}) = T(\mathbf{u}) + T(\mathbf{v}).$$

On the unit circle, $\mathbf{e}_1$ is at angle 0 and goes to the point at angle φ; $\mathbf{e}_2$ is at angle 90° and goes to the point at angle $90° + \varphi$:

$$T(\mathbf{e}_1) = \begin{bmatrix}\cos\varphi\\ \sin\varphi\end{bmatrix}, \qquad T(\mathbf{e}_2) = \begin{bmatrix}\cos(90°+\varphi)\\ \sin(90°+\varphi)\end{bmatrix} = \begin{bmatrix}-\sin\varphi\\ \cos\varphi\end{bmatrix}, \qquad A = \begin{bmatrix}\cos\varphi & -\sin\varphi\\ \sin\varphi & \cos\varphi\end{bmatrix}.$$



Rotation by φ (drawn with $\varphi = 30°$). The images of $\mathbf{e}_1$ and $\mathbf{e}_2$ are the columns of the rotation matrix.

Composition. If $S\colon \mathbb{R}^n \to \mathbb{R}^k$ and $T\colon \mathbb{R}^k \to \mathbb{R}^m$ are linear, then "first S, then T", $\mathbf{x} \mapsto T(S(\mathbf{x}))$, is linear:

$$T(S(c\mathbf{u}+d\mathbf{v})) = T(cS(\mathbf{u}) + dS(\mathbf{v})) = cT(S(\mathbf{u})) + dT(S(\mathbf{v})),$$

by $T(c\mathbf{u}+d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ (properties of linear transformations, Lay 1.8), used for S and then for T. So by Theorem 10 its standard matrix has columns $T(S(\mathbf{e}_j))$: follow each $\mathbf{e}_j$ through S and then through T.

Standard matrices on $\mathbb{R}^2$, each found by asking where $\mathbf{e}_1$ and $\mathbf{e}_2$ go:

transformation	$T(\mathbf{e}_1),\ T(\mathbf{e}_2)$	matrix
reflection through the x_1-axis	$(1,0),\ (0,-1)$	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$
reflection through the x_2-axis	$(-1,0),\ (0,1)$	$\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$
reflection through the line $x_2 = x_1$	$(0,1),\ (1,0)$	$\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$
reflection through the line $x_2 = -x_1$	$(0,-1),\ (-1,0)$	$\begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix}$
reflection through the origin	$(-1,0),\ (0,-1)$	$\begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}$
horizontal contraction/expansion by $k > 0$	$(k,0),\ (0,1)$	$\begin{bmatrix}k & 0\\ 0 & 1\end{bmatrix}$
vertical contraction/expansion by $k > 0$	$(1,0),\ (0,k)$	$\begin{bmatrix}1 & 0\\ 0 & k\end{bmatrix}$
horizontal shear by k	$(1,0),\ (k,1)$	$\begin{bmatrix}1 & k\\ 0 & 1\end{bmatrix}$
vertical shear by k	$(1,k),\ (0,1)$	$\begin{bmatrix}1 & 0\\ k & 1\end{bmatrix}$
projection onto the x_1-axis	$(1,0),\ (0,0)$	$\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$
projection onto the x_2-axis	$(0,0),\ (0,1)$	$\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$

Definition. $T\colon \mathbb{R}^n \to \mathbb{R}^m$ is **onto** $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at least one* $\mathbf{x}$ in $\mathbb{R}^n$. It is **one-to-one** if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of *at most one* $\mathbf{x}$.

Theorem (One-to-one means only $\mathbf{0}$ goes to $\mathbf{0}$ (Lay 1.9, Thm. 11))
A linear transformation T is one-to-one if and only if $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Proof. 1. **Assume T is one-to-one.** $T(\mathbf{0}) = \mathbf{0}$, and at most one vector goes to $\mathbf{0}$, so $\mathbf{0}$ is the only solution of $T(\mathbf{x}) = \mathbf{0}$.

2. **Assume only $\mathbf{0}$ goes to $\mathbf{0}$**, and let $T(\mathbf{u}) = T(\mathbf{v})$. By linearity $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$, so $\mathbf{u} - \mathbf{v} = \mathbf{0}$ and $\mathbf{u} = \mathbf{v}$. So each $\mathbf{b}$ is the image of at most one vector.

□

Theorem (Onto and one-to-one from the standard matrix (Lay 1.9, Thm. 12))
Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be linear with standard matrix A.

(a) T is onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$, that is, A has a pivot in every row.

(b) T is one-to-one if and only if the columns of A are linearly independent, that is, A has a pivot in every column.

Proof. 1. (a) T is onto exactly when $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$. By (W2) this says every $\mathbf{b}$ is a combination of the columns, that is, the columns span $\mathbb{R}^m$; by (W3) this is a pivot in every row.

2. (b) By Theorem 11, T is one-to-one exactly when $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. By definition this is independence of the columns, and by (W5) it says $A\mathbf{x} = \mathbf{0}$ has no free variable, that is, A has a pivot in every column.

□

Size check. An $m \times n$ matrix has at most m and at most n pivots (W6). If $n > m$, some column has no pivot, so T is not one-to-one. If $n < m$, some row has no pivot, so T is not onto. If $n = m$, a pivot in every column is the same as a pivot in every row (W6), so T is one-to-one exactly when it is onto.

The question answered. Yes, every linear T is $\mathbf{x} \mapsto A\mathbf{x}$ for its standard matrix, whose columns are the images $T(\mathbf{e}_j)$. For the rotation by $30°$, $\cos 30° = \frac{\sqrt{3}}{2}$ and $\sin 30° = \frac{1}{2}$, so

$$A = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}.$$

Whether every $\mathbf{b}$ is an image (onto) and of only one $\mathbf{x}$ (one-to-one) is read from the pivots of A (Theorem 12). The rotation keeps every distance from $\mathbf{0}$, so it sends only $\mathbf{0}$ to $\mathbf{0}$; by Theorem 11 it is one-to-one, and since A is square it is also onto.

Shortcuts

- **Standard matrix**: column j is $T(\mathbf{e}_j)$; for a formula, row i lists the coefficients of entry i. Reason: Theorem 10 and the row-vector rule. Instance: the worked example.
- **Composition of geometric moves**: track $\mathbf{e}_1$ and $\mathbf{e}_2$ through the moves in order. Reason: the composition is linear, so Theorem 10 applies. Instance: rotating by $90°$ twice sends $\mathbf{e}_1 \to \mathbf{e}_2 \to -\mathbf{e}_1$ and $\mathbf{e}_2 \to -\mathbf{e}_1 \to -\mathbf{e}_2$, so the matrix is $-I$, the rotation by $180°$.
- **One row reduction of A answers both questions**: a pivot in every row means onto; a pivot in every column means one-to-one. Reason: Theorem 12. Instance: the matrix of the 1.8 worked example has pivots in both columns but not in row 3, so it is one-to-one and not onto; this is why $\mathbf{b}$ had exactly one preimage and $\mathbf{c}$ had none.
- **Size check**: more columns than rows means not one-to-one; more rows than columns means not onto; square means one-to-one exactly when onto. Reason: the size check above. Instance: a linear map $\mathbb{R}^3 \to \mathbb{R}^2$ sends some nonzero vector to $\mathbf{0}$.

Traps

- "Onto" and "one-to-one" use different pivot conditions (rows versus columns). The matrix below has a pivot in every column (one-to-one) but not in every row (not onto $\mathbb{R}^3$):

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

- The minus sign of the counterclockwise rotation matrix is in the top right. With it in the bottom left, $\mathbf{e}_1$ goes to $(\cos\varphi, -\sin\varphi)$, which is the clockwise rotation.
- The order of a composition matters. Reflecting through the x_1-axis and then rotating by 90° sends $\mathbf{e}_1 \to \mathbf{e}_1 \to \mathbf{e}_2$; rotating first and then reflecting sends $\mathbf{e}_1 \to \mathbf{e}_2 \to -\mathbf{e}_2$.

§3 Homework this week

HW2, Sections 1.7–1.9, and HW3, Section 1.8: 41, 44.

Section 1.7.

- **#1, #6** (independent?). Row reduce the matrix with these columns; independent exactly when every column has a pivot.
- **#10** (span membership and dependence with h). Notice how $\mathbf{v}_1$ and $\mathbf{v}_2$ are related; then row reduce for (a) and use that relation for (b).
- **#13** (h making the set dependent). Row reduce with h as a letter and find when a column loses its pivot (as in P2).
- **#21** (wording of independence). Compare with the definition: *only* the trivial solution.
- **#24** (fewer vectors than entries). Theorem 8: check its hypothesis.
- **#26** (adding $\mathbf{z}$ from $\mathrm{Span}\{\mathbf{x}, \mathbf{y}\}$). Theorem 7.
- **#33, #36** (pivot columns of a matrix with independent columns). Independent columns means a pivot in every column.
- **#38** (nontrivial solution by inspection). Rewrite the given column relation as a relation equal to $\mathbf{0}$ and read off $\mathbf{x}$ (third shortcut of 1.7).

Section 1.8.

- **#2** (images). Compute $A\mathbf{u}$ and $A\mathbf{v}$; note that A is a multiple of I.
- **#5, #11** (find $\mathbf{x}$ with $T(\mathbf{x}) = \mathbf{b}$; is $\mathbf{b}$ in the range?). Row reduce $[\,A \;\; \mathbf{b}\,]$; uniqueness depends on free variables (worked example).
- **#7** (sizes of domain and codomain). $A\mathbf{x}$ needs as many entries in $\mathbf{x}$ as A has columns, and has as many entries as A has rows.
- **#17** (images of $3\mathbf{u}$, $2\mathbf{v}$, $3\mathbf{u} + 2\mathbf{v}$). $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$.
- **#24, #26** (codomain versus range; existence versus uniqueness). The definitions in 1.8.
- **#29** (linearity test with $c\mathbf{u} + d\mathbf{v}$). The theorem on properties of linear transformations, both directions.
- **#31** (reflection sketches). Draw $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$, $c\mathbf{u}$ and their reflections.
- **#41** (a map with a constant term). Check $T(\mathbf{0})$.
- **#44** (projection onto the x_1x_3-plane). Write it as $\mathbf{x} \mapsto A\mathbf{x}$, or check the two rules from the formula.

Section 1.9.

- **#2, #6** (standard matrix from the images of $\mathbf{e}_j$ or from a description). Theorem 10: the columns are $T(\mathbf{e}_j)$; for #6 use the shear row of the table.
- **#13** (sketch $T(2,1)$). $(2,1) = 2\mathbf{e}_1 + \mathbf{e}_2$ and superposition.
- **#15, #19** (matrix from a formula). Row i lists the coefficients of entry i (worked example).

- **#21** (find $\mathbf{x}$ with $T(\mathbf{x}) = (3, 8)$). Write the standard matrix and row reduce.
- **#23, #25** (determined by $T(\mathbf{e}_j)$; rotations linear). Theorem 10; the rotation paragraph.
- **#32, #33** (onto, one-to-one). Theorem 12 and the size check.

§4 Problem set

P1. Are the columns of the matrix below linearly independent?

$$\begin{bmatrix} 1 & -3 & 2 \\ 3 & -8 & 5 \\ -2 & 7 & -3 \end{bmatrix}$$

P2. For which h are $(1, 2, -1)$, $(2, 5, 1)$, $(1, 4, h)$ linearly dependent? For that h, give a dependence relation.

P3. Decide by inspection, with a reason: (a) $(1, 2), (3, 1), (0, 5)$; (b) $(2, -4, 6), (-1, 2, -3)$; (c) $(1, 0, 2), (0, 0, 0), (5, 1, 1)$; (d) $(1, 2, 0), (0, 1, 3)$.

P4. $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ first applies the horizontal shear by 2 and then reflects through the x_2-axis. Find its standard matrix. Is T one-to-one? Onto?

P5. Let $T(\mathbf{x}) = A\mathbf{x}$ with A below. Find all $\mathbf{x}$ with $T(\mathbf{x}) = (-1, 2, 0)$. Is T one-to-one? Onto?

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 2 & -3 & 5 \end{bmatrix}$$

P6. Is $T(x_1, x_2) = (x_1 + x_2,\ 2x_1 - 1)$ linear? Is $S(x_1, x_2) = (x_2,\ 3x_1 - x_2,\ 0)$?

P7. $T\colon \mathbb{R}^4 \to \mathbb{R}^3$ has the standard matrix below. Is T onto? One-to-one?

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 5 & 1 & 7 \\ 1 & 3 & 2 & 5 \end{bmatrix}$$

P8. $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ is linear with $T(\mathbf{e}_1) = (2, -1, 3)$ and $T(\mathbf{e}_2) = (0, 4, 1)$. Find $T(3, -2)$ and the standard matrix of T.